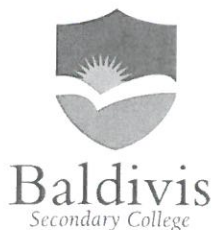


Name: _____

Sol Options

Date: _____

**Subject - Year 8 Mathematics****Investigation 1, 2018****Topic – Perimeter & Area****In class component**

$\begin{array}{r} \hline 42 \\ \hline \end{array}$
= ____ %

— 1/2 in correct units of measurement.

Total Time:	60 minutes	
Equipment:	Scientific Calculator, Take home component	

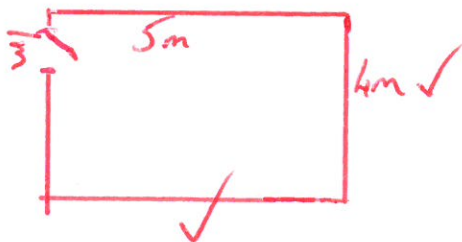
Date out: Week ____ Date ____ **Date Due:** Week ____ Date ____

AIM: In this assessment, you will be investigating area and perimeter for simple and composite shapes.

1. Perimeter of a rectangle [7 marks: 1, 1, 2, 3]

Miranda's bedroom is 5m x 4m x 2.5m high, with a doorway 1m wide and 2 m high in one of the 4m sides.

a) Sketch a floor plan of Miranda's room



b) Add the appropriate dimensions to the floor plan of Miranda's room.

c) Calculate the perimeter of the room so that Miranda can fit cornice material where the ceiling and walls meet.

$$5 + 5 + 4 + 4 = 18m$$

d) Calculate the amount of skirting board needed if Miranda is to fit skirting board where the walls and floor join. (There is no skirting where the door is)

$$5 + 5 + 4 + 3 = 17m$$

(4-1)

2. Perimeter of a Square [4 marks: 2, 2]

Miranda decides to add a square mat.

- a) What is the length of the largest mat to fit the room?

4m ✓✓

- b) Miranda wishes to add an extra strip around the perimeter to protect the edge of the carpet. How much edging material does Miranda need?

4 x 4 = 16m. ✓✓

3 Tiled rectangles [6 marks: 2, 1, 1, 1, 1]

Miranda has a very small kitchen 2m x 3m x 2.5m high and wishes to tile the floor using 200mm square tiles.

- a) What is the area of each tile?

(i) in mm^2 $200 \times 200 = 40\,000 \text{ mm}^2$ ✓

(ii) in m^2 $40,000 \div 1\,000\,000 = 0.04 \text{ m}^2$ ✓

- b) How many tiles are needed along the long wall?

$3000 \text{ mm} \div 200 \text{ mm} = 15$ ✓

- c) How many tiles are needed along the short wall?

- d) How many tiles are needed altogether?

$2000 \text{ mm} \div 200 \text{ mm} = 10$ ✓
 $15 \times 10 = 150$ ✓

- e) What is the total tiled area?

$2 \text{ m} \times 3 \text{ m} = 6 \text{ m}^2$ ✓

4 Area of triangular carpet [3 marks: 2, 1]

Miranda decided to try a triangular carpet in her 5m x 4m bedroom.

- a) What are the height and base of the maximum size of carpet that she could use?

Height 5 m or 4 m ✓

Base 4 m or 5 m. ✓

- b) What is the area of this carpet?

10 m^2

$\frac{1}{2} \times 4 \times 5 = 10$
 $\frac{1}{2} \times 5 \times 4 = 10$ ✓

5 Area of rectangular carpet [3 marks: 2, 1]

Miranda decided to try a rectangular carpet in the bedroom.

- a) What are the maximum length and width of carpet that she should use?

Length 5 m or 4 ✓

Width 4 m or 5 ✓

- b) What is the area of this carpet?

$$5 \times 4 = 20$$

$$4 \times 5 = 20$$

$$20\text{m}^2$$

6 Area of a Square carpet [2 Marks]

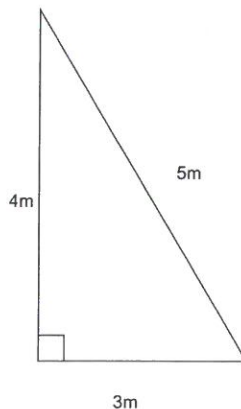
Miranda wanted to try the square carpet used previously in the perimeter question (Question 2)

What is the area of the ~~circular~~ ^{square} carpet?

$$4 \times 4 = 16\text{m}^2$$

7 Area of right angle triangle [3 marks]

Miranda wants a shade sail as shown below

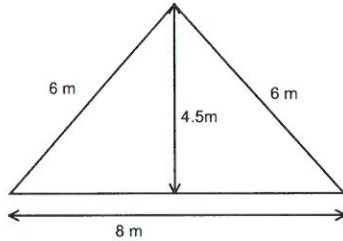


$$\begin{aligned} A &= \frac{1}{2}bh. \quad \checkmark \\ &= \frac{1}{2} \times 3 \times 4 \quad \checkmark \\ &= 6\text{m}^2 \quad \checkmark \end{aligned}$$

Calculate area of the material need to make the shade sail?

8. Area of non-right triangle [3 marks]

Miranda decides to use a design like the one below.



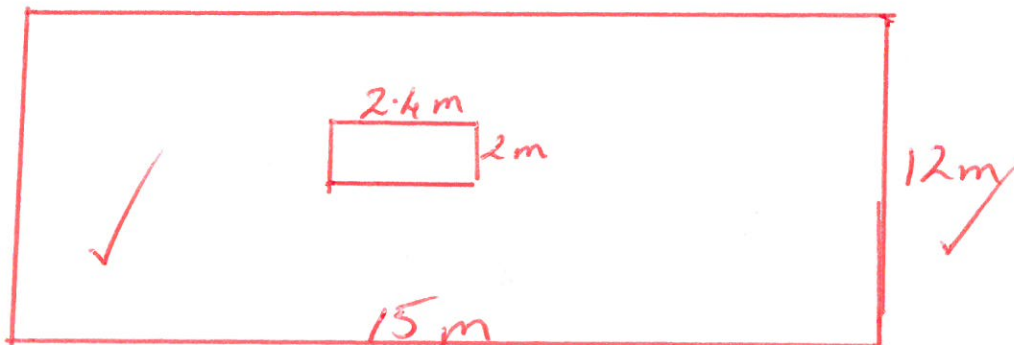
Calculate the area of material used.

$$\begin{aligned}
 A &= \frac{1}{2}bh. \checkmark \\
 &= \frac{1}{2} \times 8 \times 4.5 \checkmark \\
 &= 18m^2 \checkmark
 \end{aligned}$$

9. Area of rectangle with rectangle removed [6 marks: 2, 1, 1, 1, 1]

Miranda decides to turf the back yard which is 15m x 12m and includes a fish pond in the centre measuring 2.4m x 2m

a) Draw a sketch plan of the yard



b) Mark all dimensions you have been given on your plan. $\checkmark$

c) Calculate the area of the yard?

$$15 \times 12 = 180m^2 \checkmark$$

d) Calculate the area of the pond?

$$2.4 \times 2 = 4.8m^2 \checkmark$$

e) Calculate the area of turf needed?

$$180m^2 - 4.8m^2 = 175.2m^2 \checkmark$$

10. Rectangle with Trapezium removed [3 marks: 2,1]

Miranda decides to have a trapezoidal fish pond with a top, bottom and height of 2.4 m, 3.2m and 2m respectively instead.

(The back yard is still 15m x 12m.)

- a) Calculate the area of the pond

$$\begin{aligned} A &= h \left(\frac{a+b}{2} \right) \\ &= 2 \left(\frac{2.4 + 3.2}{2} \right) \checkmark \\ &= 5.6 \text{ m}^2 \checkmark \end{aligned}$$

- b) Calculate the area of lawn left after the pond is dug out.

$$180 - 5.6 = 174.4 \text{ m}^2$$

